

CLEVELAND HARBOR, OH, USA

WMO: 997692

Lat: 41.542N

Lon: 81.637W

Elev: 210

StdP: 98.83

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 2	(b) -15.3	(c) -12.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 14.5	(k) -2.8	(l) 13.0	(m) -2.6	(n) 4.0	(o) 220	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.1	(c) 30.5	(d) N/A	(e) 28.8	(f) N/A	(g) 27.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.0	(b) 11.7	(c) 10.4	(d) DB	(e) -17.7	(f) 33.4	(g) 4.9	(h) 1.1	(i) -21.3	(j) 34.2	(k) -24.1	(l) 34.8	(m) -26.9	(n) 35.4	(o) -30.4	(p) 36.1	(q) N/A
(a) 13.0	(b) 11.7	(c) 10.4	(d) WB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.6	-2.5	-2.6	2.3	9.3	15.7	20.2	22.6	21.9	18.8	12.8	6.2	2.1
	DBStd	10.09	6.22	6.07	5.59	5.18	4.78	3.58	2.97	2.64	3.73	4.18	4.61	4.97
	HDD10.0	1471	389	354	249	74	6	0	0	0	0	18	133	248
	HDD18.3	3224	644	586	497	275	113	20	2	3	38	179	365	502
	CDD10.0	1703	3	2	10	54	181	306	391	368	263	104	18	4
	CDD18.3	416	0	0	0	5	30	77	134	112	51	7	0	0
	CDH23.3	2288	0	0	0	48	166	417	830	527	274	26	0	0
	CDH26.7	480	0	0	0	7	33	102	171	97	69	1	0	0
(14) Wind	WSAvg	3.9	4.6	4.4	4.5	4.6	3.4	3.1	3.0	3.0	3.4	4.2	4.5	4.5
(15) Precipitation	PrecAvg	1023	76	58	71	89	95	97	106	90	100	88	73	80
	PrecMax	1334	154	139	153	147	186	202	235	199	228	172	127	180
	PrecMin	716	37	24	16	44	35	21	32	32	30	37	31	40
	PrecStd	145	29	25	28	28	44	42	48	35	44	31	26	28
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.0	18.6	21.4	27.9	29.9	32.0	32.0	32.2	31.8	26.0	19.0	17.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	13.4	12.4	18.1	24.9	27.3	29.4	30.0	29.1	28.8	23.6	17.2	14.9
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	9.7	8.7	15.2	21.7	25.1	27.5	28.5	27.1	26.3	21.7	15.4	12.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	6.1	5.1	11.5	18.8	22.9	25.7	27.3	25.9	24.5	19.6	13.5	9.5
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(35) Mean Daily Temperature Range	5% DB	MDBR	6.4	6.5	7.1	8.1	7.8	6.6	6.1	6.0	6.8	7.0	6.3	5.7
		MCDBR	10.2	11.1	12.7	13.8	11.2	9.6	8.0	7.9	9.3	9.5	9.5	9.2
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(40) Clear-Sky Solar Irradiance	taub		0.306	0.335	0.359	0.397	0.433	0.451	0.451	0.436	0.388	0.369	0.328	0.309
	taud		2.402	2.278	2.293	2.275	2.208	2.201	2.224	2.257	2.365	2.397	2.488	2.478
	Ebn at Noon		842	863	885	873	846	829	825	827	845	818	814	811
	Edh at Noon		80	106	118	129	141	143	138	130	109	94	74	70
(44) All-Sky Solar Radiation	RadAvg		1.36	2.23	3.26	4.46	5.37	5.96	5.98	5.28	4.20	2.67	1.68	1.09
	RadStd		0.12	0.28	0.33	0.39	0.46	0.45	0.39	0.25	0.38	0.23	0.27	0.10

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+100	N/A

Nomenclature: See separate page